Practical 1 wine

1. First, understand what this practical does

Imagine you have a wine dataset.

This dataset contains different properties of wine, such as:

- Alcohol
- Malic Acid
- Ash
- Magnesium
- Total Phenols
- Flavanoids
- Color Intensity
- Hue

There are many columns (features) in the dataset.

Our task is to reduce these many features into a smaller number of features without losing too much important information.

This process is called Dimensionality Reduction.

2. What is PCA?

PCA stands for Principal Component Analysis.

PCA is an unsupervised machine learning algorithm used for dimensionality reduction.

In simple words:

PCA reduces the number of features in a dataset while preserving as much important information as possible.

Example:

Suppose a dataset has 13 features.

Using PCA, we can convert these 13 features into just 2 principal components while retaining a large amount of information.

This makes the dataset easier to understand and visualize.

3. Important terms you must understand

Term	**Simple meaning**
Dimensionality	Number of features or columns

Feature	An input variable, like Alcohol
PCA	Technique used to reduce features
Principal Components	New features created by PCA
Variance	How much the data values are spread out
Explained Variance	How much information is retained
Standardization	Bringing features to a similar scale
Unsupervised Learning	Learning without using target labels during training

4. Aim of the practical-the aim given in your journal is:

To use the PCA algorithm for dimensionality reduction on a wine dataset containing different measurements such as alcohol, ash and magnesium. We apply PCA to reduce the dimensions while retaining maximum variation in the data, making it easier to distinguish red and white wine.

5. Which dataset are we using?

We are using a Wine Dataset.

It contains measurements of different chemical properties of wine.

The dataset has two types of information:

- Input features (X): Alcohol, Malic Acid, Ash, Magnesium, etc.
- Target variable (y): Wine category or class.

The practical uses the features to perform PCA and then plots the wine categories using two principal components.

6. Why do we use PCA?

There are several reasons:

1. Reduces dimensions: Reduces the number of features.
2. Removes redundancy: Avoids repeated or highly correlated information.
3. Easy visualization: Makes it possible to display high-dimensional data in 2D or 3D.
4. Improves efficiency: Reduces computational complexity.
5. Preserves information: Tries to retain maximum variance in fewer components.

Viva question: What is Pandas?

Answer: Pandas is a Python library used for data manipulation and analysis.

What is Matplotlib?

Answer: Matplotlib is a Python library used for data visualization and plotting graphs.

Step 2: Load the dataset

We load the wine CSV file into a DataFrame.

What is CSV?

CSV stands for Comma Separated Values. It is a file format used to store tabular data.

What is DataFrame?

A DataFrame is a two-dimensional data structure containing rows and columns.

After loading the dataset, we display its first five rows.

Why?

To understand what data is present in our dataset.

Step 3: Check missing values

We check whether the dataset contains any null or missing values.

What are missing values?

Missing values are empty or unavailable values in a dataset.

Why do we check them?

Because missing values can affect data analysis and processing.

Your journal shows the missing-value count for each column.

Step 4: Separate X and y

This is an important concept for your viva.

- X: Independent variables or input features.
- y: Dependent variable or target variable.

For example:

X contains Alcohol, Ash, Magnesium and other chemical measurements.

y contains the wine category.

We separate them because PCA is applied to the input features, not the target column.

Step 5: Standardization

This is a very important step.

Different features have different measurement scales.

For example:

- Alcohol may have values around 10.
- Magnesium may have values around 100.
- Other features may have much smaller values.

If we directly apply PCA, features with larger scales may influence the result more.

Therefore, we use StandardScaler.

It converts the features into a common scale, generally with:

- Mean = 0
- Standard deviation = 1

Viva question: Why is standardization necessary in PCA?

Answer: Standardization is necessary because PCA is affected by the scale of features. It brings all features to a common scale.

Step 6: Apply PCA

Now we apply the PCA algorithm.

PCA identifies new variables called principal components.

Remember these points:

- PC1 = First Principal Component.
- PC2 = Second Principal Component.
- PC1 captures the maximum variance.
- PC2 captures the next highest variance.
- Both components are perpendicular (orthogonal) to each other.

In your practical, PCA is used to convert the original features into principal components.

Step 7: Explained Variance Ratio

Your journal displays a graph called Explained Variance Ratio on page 12

What does this graph show?

It shows how much information is retained when we increase the number of principal components.

For example, if the cumulative explained variance is 0.80, it means 80% of the total variance is retained.

Viva question: What is explained variance ratio?

Answer: It tells us how much information or variance is captured by each principal component.

Step 8: Final graph – Red Wine vs White Wine

This is the most important output of your practical.

The graph contains:

- X-axis: Principal Component 1.
- Y-axis: Principal Component 2.
- Red dots: Red wine.
- Blue dots: White wine.

Each dot represents one wine sample.

We use this graph to visualize how the two wine categories are distributed after dimensionality reduction.

What is a scatter plot?

A scatter plot is a graph that represents data points using dots to show the relationship between two variables.

8. Conclusion of the practical

We successfully implemented PCA for dimensionality reduction on the wine dataset. PCA converted the original features into principal components while retaining important information. The scatter plot helped visualize and distinguish the red and white wine categories.

Practical 2

1. Aim of the practical

As given in your journal:

To use the Diabetes dataset from UCI and Pima Indians Diabetes dataset to perform:

1. Univariate Analysis.
2. Bivariate Analysis.
3. Multiple Regression Analysis.
4. Compare the results of the analysis for the two datasets.

In simple words: We are studying diabetes-related data using statistical methods and regression techniques to understand patterns and relationships between different variables.

2. First, understand the dataset

We use the Pima Indians Diabetes Dataset.

This dataset contains information about patients, such as:

Important columns in the dataset

Column	**Meaning**
Pregnancies	Number of times pregnant
Glucose	Blood glucose level
BloodPressure	Blood pressure
SkinThickness	Skin thickness measurement
Insulin	Insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes-related family history score
Age	Age of patient
Outcome	Diabetes result

Outcome values

- 0 = Patient does not have diabetes.
- 1 = Patient has diabetes.

Your journal shows the dataset has:

- 768 rows (observations).
- 9 columns (attributes).

Viva question: What is a dataset?

Answer: A dataset is a collection of related data arranged in rows and columns.

3. What is Descriptive Statistics?

Descriptive statistics is a method of analysing and summarizing data.

It helps us understand the important characteristics of a dataset.

For example, suppose we have the ages of 5 patients:

20, 25, 30, 35, 40.

Using descriptive statistics, we can calculate the average age, middle value, most frequent value and spread of the data.

There are two main types of descriptive statistics:

1. Measures of Central Tendency.
2. Measures of Variability.

4. Measures of Central Tendency

These measures tell us the central or average value of the dataset.

There are three important measures:

A. Mean

Mean means average.

Formula:

\[Mean=\frac{\text{Sum of all values}}{\text{Number of values}} \]

Example: 10, 20, 30

Mean = (10 + 20 + 30) / 3 = 20.

B. Median

Median is the middle value of the data after arranging it in ascending order.

Example: 10, 20, 30, 40, 50

Median = 30.

C. Mode

Mode is the value that occurs most frequently.

Example: 10, 20, 20, 30, 40

Mode = 20.

Remember this for viva

- Mean = Average.
- Median = Middle value.

- Mode = Most repeated value.

5. Measures of Variability

These measures tell us how much the data is spread out or scattered.

There are five important measures in your practical.

Measure	Meaning
Variance	Measures how far values are spread from the mean
Standard Deviation	Measures the amount of variation in data
Skewness	Measures the asymmetry of data
Kurtosis	Measures the shape and tail characteristics of distribution
Frequency	Number of times a value occurs

Let's understand them one by one.

1. Variance: It tells us how much the values differ from the average.

2. Standard Deviation: It tells us how much the values generally vary from the mean. A higher standard deviation means more spread.

3. Skewness: It tells us whether the data distribution is symmetrical or tilted towards one side.

- Positive skewness = Right-tailed distribution.
- Negative skewness = Left-tailed distribution.
- Zero skewness = Symmetrical distribution.

4. Kurtosis: It describes the tail heaviness and peakedness of a distribution compared with a normal distribution.

5. Frequency: It tells us how many times a particular value appears.

6. Types of Analysis performed in this practical

This is very important because your examiner may ask what the difference is.

Part A: Univariate Analysis

Univariate means analysing only ONE variable at a time.

Example: We analyse only the Age column of patients.

We calculate:

- Frequency
- Mean
- Median
- Mode

- Variance
- Standard Deviation
- Skewness
- Kurtosis

Example: Finding the average age of all patients.

Part B: Bivariate Analysis

Bivariate means analysing TWO variables together to understand their relationship.

Example:

- Glucose and Outcome.
- BMI and Outcome.
- Age and Outcome.

In your practical, we perform:

1. Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values.

Example: Predicting a patient's numerical measurement based on other features.

2. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification.

Example: Predicting whether a patient has diabetes or does not have diabetes.

The output is:

- 0 = No diabetes.
- 1 = Diabetes.

Important difference:

Linear Regression	**Logistic Regression**
Used for regression	Used for classification
Predicts continuous values	Predicts class/category
Example: Predicting price	Example: Diabetes prediction

Part C: Multiple Regression Analysis

Multiple Regression means predicting one dependent variable using multiple independent variables.

For example:

We use:

- Pregnancies
- Glucose
- BloodPressure
- Insulin
- BMI
- Age

To predict the Outcome variable in your practical.

Remember:

- Independent variables = Input variables (X).
- Dependent variable = Output variable (y).

7. Correlation Heatmap and Scatter Plot

Your journal also contains two important visualizations.

A. Correlation Heatmap

A heatmap shows the correlation between different variables using colours.

Correlation tells us how two variables are related.

- +1 = Perfect positive correlation.
- -1 = Perfect negative correlation.
- 0 = No linear correlation.

B. Scatter Plot Matrix

Come creare grafici su dati multidimensionali | Intelligenza Artificiale Italia Blog

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A scatter plot matrix shows the relationship between multiple variables using graphs.

It helps us identify patterns and relationships in the dataset.

8. What is R-squared?

Your practical's multiple regression output contains an R-squared value.

R-squared tells us how much variation in the dependent variable is explained by the independent variables in the model.

Your journal shows an R-squared value of approximately 0.303.

This means the model explains approximately 30.3% of the variation in the Outcome variable.

Practical 2 workflow

1. Load dataset

Read the diabetes CSV file using Pandas.

2. Explore dataset

Check rows, columns and basic information.

3. Descriptive statistics

Calculate mean, median, mode, variance and standard deviation.

4. Univariate analysis

Analyse one variable at a time.

5. Bivariate analysis

Perform Linear Regression and Logistic Regression.

6. Multiple Regression

Analyse multiple independent variables together.

7. Visualization

Display correlation heatmap and scatter plot matrix.

L Practical 2 – Viva test

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1. What is descriptive statistics?

It is a method of collecting, analysing and summarizing data.

2. What is univariate analysis?

Analysis of only one variable at a time.

3. What is bivariate analysis?

Analysis of two variables to understand their relationship.

4. What is mean?

Mean is the average of all values.

5. What is median?

Median is the middle value of arranged data.

6. What is mode?

Mode is the most frequently occurring value.

7. What is variance?

Variance measures the spread of data from its mean.

8. What is standard deviation?

It measures how much data values vary from the mean.

9. What is skewness?

Skewness measures the asymmetry of a data distribution.

10. What is kurtosis?

Kurtosis describes the tail heaviness and shape of a distribution.

11. What is Linear Regression?

A supervised learning algorithm used to predict continuous values.

12. What is Logistic Regression?

A supervised learning algorithm used for classification.

13. What is Multiple Regression?

A regression technique that uses multiple independent variables to predict one dependent variable.

14. What is correlation?

Correlation measures the relationship between two variables.

15. What is a heatmap?

A graphical representation of data using different colours.

16. What is R-squared?

It measures the proportion of variation in the dependent variable explained by the regression model.

17. What is the outcome column?

It represents the diabetes result: 0 for non-diabetic and 1 for diabetic.

Hide answers

11. How to explain this practical to the examiner

Learn this paragraph. You can speak it directly in your practical exam.

"My practical is based on Descriptive Statistics and Regression Analysis using the Diabetes dataset. First, we load the dataset and explore its different columns. Then we perform Univariate Analysis to calculate mean, median, mode, variance, standard deviation, skewness and kurtosis. After that, we perform Bivariate Analysis using Linear and Logistic Regression. We also perform Multiple Regression Analysis and visualize relationships using a correlation heatmap and scatter plot matrix."

12. Conclusion

We successfully performed descriptive statistical analysis on the diabetes dataset. We calculated different statistical measures and applied Linear Regression, Logistic Regression and Multiple Regression to understand relationships between variables. We also visualized the data using a correlation heatmap and scatter plot matrix.

Practical 3: Classification Analysis using SVM

Confirmed Practical 3

1. Aim of the practical

To implement the Support Vector Machine (SVM) algorithm for classifying images of handwritten digits into their respective numerical classes from 0 to 9.

In simple words, we train a computer to recognize handwritten numbers from images.

2. What is classification?

Classification is a type of supervised machine learning in which a model learns from labelled data and predicts the category of new data.

For example:

- Input: An image of a handwritten digit.
- Output: The number predicted by the computer, such as 5.

There are 10 possible classes: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

MNIST Handwritten Digits [IMAGE] | EurekAlert! Science News Releases

Optical Recognition of Handwritten Digits using Scikit-Learn | by Ayush Kumar | Medium

Azure Machine Learning Service for Kubernetes Architects: Deploy Your First Model on AKS with AZ CLI v2 | by Joseph Masengesho | Medium

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3. What is SVM?

SVM stands for Support Vector Machine.

It is a supervised machine learning algorithm used mainly for classification and also for regression.

SVM works by finding the best possible boundary, called a hyperplane, that separates different classes of data.

For example, imagine two groups of points. SVM tries to draw a boundary between them so that the groups are separated as clearly as possible.

4. Dataset used in your actual practical

I checked your uploaded practical journal. It uses Scikit-learn's built-in Digits dataset, not a separate downloaded image folder.

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Important details:

Dataset detail	**Value**
Total handwritten digit images	1,797
Image size	8 × 8 pixels
Number of features per image	64
Number of classes	10 (0–9)

Pixel value range	0 to 16

What does 8 × 8 pixels mean?

Each digit image is made up of 8 rows and 8 columns of tiny squares called pixels.

So, \(8 \times 8 = 64\) pixels. Each pixel has a numerical brightness value. The 64 pixel values become the input features for the machine learning model.

- X contains the pixel values (input/features).
- y contains the correct digit labels (output/target).

For example, if an image shows the handwritten digit 7, its label in y is 7.

5. Important SVM terms

1. Hyperplane: A decision boundary that separates different classes. In this practical, SVM uses these boundaries to distinguish handwritten digits.

2. Support vectors: The data points closest to the decision boundary. They help determine where the boundary is placed.

3. Kernel: A function that helps SVM separate data when a straight boundary is not sufficient.

Your practical compares four kernels:

Kernel	**Simple meaning**
Linear	Uses a straight decision boundary
Polynomial	Uses a curved or more complex boundary
RBF	Handles complex, non-linear patterns
Sigmoid	Uses a boundary based on a sigmoid function

4. Feature scaling: Brings the input features to a comparable scale. It is important because SVM is sensitive to the scale of input data.

6. Steps performed in your practical

The following sequence follows the practical journal.

1. Import required libraries

Libraries such as NumPy, Pandas, Matplotlib, Seaborn and Scikit-learn are imported for data handling, visualization, SVM training and evaluation.

2. Load the Digits dataset

The dataset is loaded using load_digits(). The input features and target labels are separated into X and y.

3. Display sample digit images

Sample images from all 10 classes are displayed. This helps us see how handwritten digits look in the dataset.

4. Split the dataset

The data is divided into:

- 80% training data — used to teach the model.
- 20% testing data — used to check its performance.

stratify=y helps maintain the proportion of each digit class in both sets.

5. Apply feature scaling

StandardScaler standardizes the pixel features so that they are on a comparable scale.

6. Train the SVM model

The model learns patterns from the training data. Your journal first uses an RBF kernel as the baseline model.

7. Compare different kernels

Linear, Polynomial, RBF and Sigmoid kernels are compared to understand which works better for the dataset.

8. Find the best parameters

GridSearchCV tests different parameter combinations and uses cross-validation to select a good model configuration.

9. Evaluate the model

The final model is tested using accuracy, precision, recall, F1-score and a confusion matrix.

7. Understanding the output of your practical

Your journal gives these results:

Baseline SVM model

Output	**Result**
Training time	0.030 seconds
Test accuracy	97.50%
Support vectors used	746

The baseline model uses the RBF kernel.

After GridSearchCV

The best parameters found were:

Parameter	**Best value**
Kernel	Polynomial (poly)
C	10

Degree	3
Gamma	scale
Best cross-validation accuracy	98.26%

The final test results are:

99.44%

Final test accuracy

Evaluation measure	**Result**
Accuracy	99.44%
Precision (macro)	99.45%
Recall (macro)	99.44%
F1-score (macro)	99.44%

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This means the final model correctly classified almost all the handwritten digit images in the test dataset.

What are these evaluation measures?

- Accuracy: How many total predictions were correct.
- Precision: Out of all the images predicted as a particular digit, how many were actually that digit?
- Recall: Out of all actual images of a digit, how many did the model correctly identify?
- F1-score: A combined measure of precision and recall.

What is a confusion matrix?

A confusion matrix is a table that compares the actual digit labels with the predicted digit labels.

- Rows represent actual digits.
- Columns represent predicted digits.
- Diagonal values represent correct predictions.
- Values outside the diagonal represent incorrect predictions.

Your practical also displays a normalised confusion matrix, which shows the proportion of correctly identified examples for each digit.

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8. Important viva questions and answers

Q1. What is the full form of SVM?

Support Vector Machine.

Q2. What is SVM?

SVM is a supervised machine learning algorithm used for classification and regression.

Q3. What is classification?

Classification is the process of predicting the category or class of given data.

Q4. Which dataset is used in this practical?

The built-in Digits dataset from the Scikit-learn library.

Q5. How many images are present in the dataset?

1,797 images.

Q6. What is the size of each image?

8 × 8 pixels.

Q7. How many classes are there?

10 classes, representing digits from 0 to 9.

Q8. What are support vectors?

Support vectors are the data points closest to the decision boundary. They help determine the position of the boundary.

Q9. What is a kernel in SVM?

A kernel helps SVM create decision boundaries for separating data, including data that cannot be separated using a straight line.

Q10. Which kernels are compared in this practical?

Linear, Polynomial, RBF and Sigmoid.

Q11. Why is feature scaling required?

It brings features to a comparable scale and helps SVM perform better.

Q12. What is GridSearchCV?

It tests different combinations of model parameters using cross-validation to find a suitable combination.

Q13. What is the accuracy of your final model?

99.44%.

Q14. What is the difference between training and testing data?

Training data is used to teach the model, while testing data is used to evaluate its performance on unseen examples.

9. What to tell the examiner

You can speak this explanation during your practical exam:

"Good morning, sir/madam. The aim of this practical is to implement Support Vector Machine for handwritten digit recognition.

We use the Scikit-learn Digits dataset, which contains 1,797 images of handwritten digits from 0 to 9. Each image is of size 8 by 8 pixels, giving 64 features.

First, we load the dataset and display sample digit images. Then we divide the dataset into training and testing data. We apply feature scaling and train the SVM classifier.

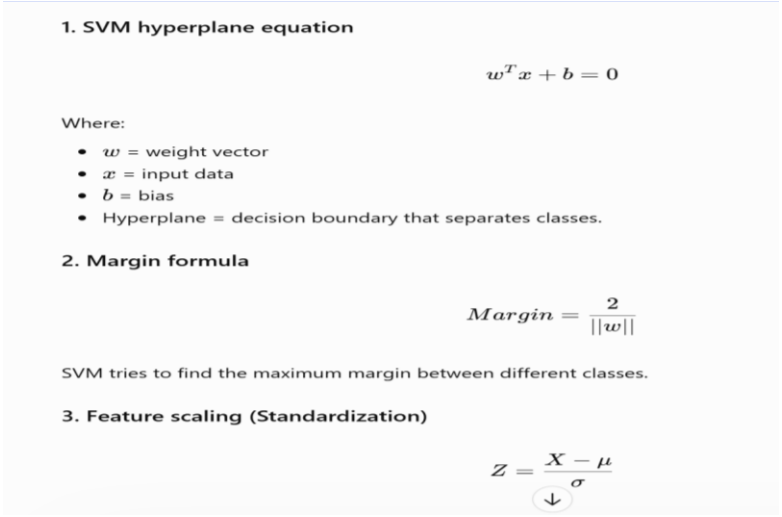
We compare different kernels such as Linear, Polynomial, RBF and Sigmoid. We also use GridSearchCV to find the best parameters.

Finally, we evaluate the model using accuracy, precision, recall, F1-score and a confusion matrix. The final model achieves 99.44% test accuracy.

Thus, we successfully classify handwritten digits using the SVM algorithm."

10. Conclusion

In this practical, we successfully implemented Support Vector Machine for handwritten digit classification. We compared different kernels and optimized the model using GridSearchCV. The final model achieved 99.44% accuracy, showing that SVM can effectively recognize handwritten digits. **Important formulas for Practical 3 (SVM)**



1. SVM hyperplane equation

$$w^T x + b = 0$$

Where:

- w = weight vector
- x = input data
- b = bias
- Hyperplane = decision boundary that separates classes.

2. Margin formula

$$Margin = \frac{2}{||w||}$$

SVM tries to find the maximum margin between different classes.

3. Feature scaling (Standardization)

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = original value
- μ = mean
- σ = standard deviation
- Z = standardized value

4. Accuracy

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \times 100$$

5. Precision

$$Precision = \frac{TP}{TP + FP}$$

6. Recall

$$Recall = \frac{TP}{TP + FN}$$

7. F1-score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Remember these terms:

Term	Meaning
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

No, there is **no single universal formula for all SVM kernels**. Each kernel has its own formula.

But the general kernel function is written as:

$$\boxed{K(x, y) = \phi(x)^T \phi(y)}$$

Here:

- $K(x, y)$ = Kernel function
- x, y = Two data points
- ϕ = Transformation of data into a higher-dimensional space

For your practical, remember these four kernel formulas:

1. Linear: $K(x, y) = x^T y$
2. Polynomial: $K(x, y) = (\gamma x^T y + r)^d$
3. RBF: $K(x, y) = e^{-\gamma ||x-y||^2}$
4. Sigmoid: $K(x, y) = \tanh(\gamma x^T y + r)$